Algebra: Parallel & Perpendicular Lines

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DIRECTIONS *Find the equation of a line parallel or perpendicular to the given line through a point.*

1 Line through (1,2) perpendicular to:

$$y = 2x + 2$$

Answer: _____

2 Line through (2,-1) parallel to:

$$y = -3x + 5$$

Answer: _____

3 Line through (0,3) perpendicular to:

$$y = \frac{1}{2}x + 2$$

Answer: _____

4 Line through (-1,4) perpendicular to:

$$y = -\frac{2}{3}x + 2$$

Answer: _____

5 Line through (3,0) parallel to:

$$y = \frac{3}{4}x + 5$$

Answer: _____

6 Line through (-2,-3) perpendicular to:

$$y = -\frac{1}{3}x + 2$$

Answer: _____

7 Line through (1,-2) perpendicular to:

$$y = 4x + 2$$

Answer: _____

8 Line through (0,5) perpendicular to:

$$y = -\frac{5}{2}x + 2$$

Answer: _____

9 Line through (-3,1) perpendicular to:

$$y = \frac{1}{4}x + 2$$

Answer:

10 Line through (2,4) parallel to:

$$y = \frac{3}{2}x + 5$$

Answer:

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Answer Key & Solutions

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TEACHER NOTES *Parallel: same slope. Perpendicular: negative reciprocal slope ($m_1 \cdot m_2 = -1$).*

1 Line through (1,2) perpendicular to:

= $y = -\frac{1}{2}x + \frac{5}{2}$

$y = 2x + 2$

2 Line through (2,-1) parallel to:

= $y = -3x + 5$

$y = -3x + 5$

3 Line through (0,3) perpendicular to:

= $y = -2x + 3$

$y = \frac{1}{2}x + 2$

4 Line through (-1,4) perpendicular to:

= $y = \frac{3}{2}x + \frac{11}{2}$

$y = -\frac{2}{3}x + 2$

5 Line through (3,0) parallel to:

= $y = \frac{3}{4}x + -\frac{9}{4}$

$y = \frac{3}{4}x + 5$

6 Line through (-2,-3) perpendicular to:

= $y = 3x + 3$

$y = -\frac{1}{3}x + 2$

7 Line through (1,-2) perpendicular to:

= $y = -\frac{1}{4}x + -\frac{7}{4}$

$y = 4x + 2$

8 Line through (0,5) perpendicular to:

= $y = \frac{2}{5}x + 5$

$y = -\frac{5}{2}x + 2$

9 Line through (-3,1) perpendicular to:

= $y = -4x + -11$

$y = \frac{1}{4}x + 2$

10 Line through (2,4) parallel to:

= $y = \frac{3}{2}x + 1$

$y = \frac{3}{2}x + 5$

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